Residential HVAC Design Package

Room-by-room load calculation (ACCA Manual J, 8th Ed. structure) · Equipment selection (ACCA Manual S) · Duct design (ACCA Manual D) · Whole-dwelling ventilation (ASHRAE 62.2)

SAMPLE DOCUMENT – Fictitious house. Design conditions, construction data and equipment performance are illustrative values, not customer data. Calculations use simplified factors that follow the structure of the ACCA manuals. Permit submittals are produced with ACCA-approved software and site design data.

Project	Sample Residence – New Construction
Address	123 Example Lane, Morristown, NJ 07960 (fictitious)
Client / builder	Example Builders LLC (fictitious)
Description	Two-story single-family home, 4 bedrooms, 2,400 sq ft conditioned floor area, slab-on-grade, vented attic. All-electric: ducted cold-climate heat pump, ERV, heat pump water heater. Air handler and all ducts inside conditioned space (open-web floor trusses between floors).
Energy code	2021 IECC (as adopted in New Jersey)
Prepared by	[Designer Name], [Company Name] – Remote HVAC & Energy Design
Date / revision	11/03/2026 · Rev. 0 – issued for permit review

24,819 Btu/h
Heating design load (incl. ventilation)

16,031 Btu/h
Total cooling load = 1.34 tons

12,711 / 3,320
Sensible / latent cooling, Btu/h

10.3 Btu/h·ft²
Heating per ft² (2,400 ft² CFA)

2-ton class
Variable-capacity cold-climate heat pump

700 cfm
Design blower airflow (heating and cooling)

16 °F
Thermal balance point · 5 kW aux heat

0.131
Design friction rate, in. w.c. per 100 ft

Plans examiner summary

Layout follows the information requested on the ACCA Residential Plans Examiner Review Form (loads, equipment, ducts).

Item	Value	Check
Design conditions		
Outdoor heating / cooling design dry bulb	8 °F / 89 °F (coincident WB 73 °F)	
Indoor heating / cooling	70 °F / 75 °F at 50 % RH	
Grains difference · daily range · elevation	35 gr/lb · M · 400 ft	
Manual J loads		
Heating load	24,819 Btu/h	
Sensible / latent / total cooling load	12,711 / 3,320 / 16,031 Btu/h	
Duct loads	0 Btu/h – air handler and all ducts inside conditioned space	
Manual S equipment		
Cooling capacity at design (total / sensible / latent)	20,000 / 14,800 / 5,200 Btu/h	125% of total load
Heat pump heating capacity at 8 °F	20,000 Btu/h	81% of heating load
Auxiliary heat	5.0 kW = 17,061 Btu/h	required ≥ 4,819 Btu/h
Manual D ducts		
Blower external static pressure / available static pressure	0.50 / 0.25 in. w.c.	
Total effective length (supply + return)	111 + 80 = 191 ft	critical run S7 (Bedroom 3)
Design friction rate	0.131 in. w.c. / 100 ft	**PASS** within 0.06–0.18
Ventilation		
ASHRAE 62.2 whole-dwelling rate	109.5 cfm required · 110 cfm provided by ERV	**PASS**

SAMPLE

1 Design data and construction

Parameter	Value	Source / remark
Location / climate zone	Morristown, NJ (Morris County) · IECC climate zone 5A	
Heating design (99 %)	8 °F outdoor · 70 °F indoor · ΔT = 62 °F	Illustrative values – take from ACCA Manual J Table 1A for the actual site
Cooling design (1 %)	89 °F DB / 73 °F WB · 75 °F indoor · ΔT = 14 °F	Illustrative values – take from ACCA Manual J Table 1A for the actual site
Grains difference	35 gr/lb	outdoor minus indoor at 50 % RH
Altitude correction factor	1.00	elevation 400 ft

Construction components

Component	Description	U / F factor	Cooling factor
Exterior walls	2x6 @ 16" o.c., R-21 cavity + R-5 continuous exterior insulation	U-0.043	CLTD 18 °F
Ceiling	Ceiling below vented attic, R-60 blown cellulose, raised-heel trusses	U-0.024	CLTD 47 °F
Windows	Double-pane low-e, argon, vinyl frame (NFRC)	U-0.27 · SHGC 0.35	PSF N/E/S/W 20/68/35/68 · IAC 0.75
Doors	Insulated fiberglass entry door	U-0.20	CLTD 18 °F
Slab	Slab-on-grade, R-10 vertical edge insulation 4 ft	F-0.52 Btu/h·ft·°F	no cooling load

Item	Value	Remark
Air leakage	2.5 ACH50	Design target, to be verified by blower door test
Natural infiltration	heating 0.156 ACH = 53 cfm · cooling 0.125 ACH = 42 cfm	ACH50 / N with N = 16 (heating), 20 (cooling); allocated by exposed wall area
Conditioned volume / floor area	20,400 ft³ / 2,400 ft²	
Occupants / internal gains	5 persons · 230 Btu/h sensible + 200 Btu/h latent each	kitchen appliances 1,200 Btu/h, study 300 Btu/h
Ventilation	Balanced ERV, continuous, dedicated ducts · 110 cfm	SRE 70% heating · SE 65% / LE 50% cooling

2 Room-by-room loads (Manual J)

Room	Floor	Area ft²	Heating Btu/h	Sens. clg Btu/h	Lat. clg Btu/h	Htg cfm	Clg cfm	Design cfm
Study	First	168	2,308	1,232	299	72	71	**72** Htg
Entry / Stair	First	224	1,566	344	61	49	20	**49** Htg
Mudroom / Laundry / Powder	First	168	2,334	622	99	72	36	**72** Htg
Living Room	First	320	3,807	1,994	537	118	115	**118** Htg
Kitchen / Dining	First	320	3,371	2,946	537	105	170	**170** Clg
Bedroom 3	Second	182	1,672	890	92	52	51	**52** Htg
Hall / Bath 2	Second	196	1,053	498	48	33	29	**33** Htg
Bedroom 4	Second	182	1,672	890	92	52	51	**52** Htg
Primary Bedroom	Second	288	2,294	1,316	116	71	76	**76** Clg
Primary Bath	Second	160	769	419	34	24	24	**24** Clg
Bedroom 2	Second	192	1,723	969	96	53	56	**56** Clg
Room totals		**2,400**	**22,568**	**12,118**	**2,012**	**700**	**700**	
Ventilation (ERV, 110 cfm)			2,251	593	1,309			
Duct gains / losses			0	0	0			
Whole-house equipment load			**24,819**	**12,711**	**3,320**			

Room airflow = room load share × blower airflow in each mode. The larger value governs the branch size (Htg / Clg). Ventilation loads are carried by the equipment, not by room airflows (ERV ducted separately).

Load components – whole house

Component	Heating Btu/h	%	Sensible cooling Btu/h	%
Walls	6,006	24%	1,744	14%
Windows	6,144	25%	–	
Doors	496	2%	144	1%
Ceiling	1,786	7%	1,354	11%
Slab edge	4,514	18%	–	
Infiltration	3,623	15%	655	5%
Ventilation (ERV)	2,251	9%	593	5%
Windows (cond.)	–		1,387	11%
Windows (solar)	–		4,185	33%
People	–		1,150	9%
Appliances	–		1,500	12%

3 Equipment selection (Manual S)

Equipment	Variable-capacity cold-climate air-source heat pump, ducted (2-ton class)
Model / AHRI	Generic example – illustrative expanded performance data (not a real product) · AHRI [AHRI reference no.]
Rated efficiency	SEER2 18.0 · HSPF2 9.5
Cooling data basis	89 °F outdoor, 75 °F / 63 °F WB entering, 700 cfm (max. capacity, installer cap setting)

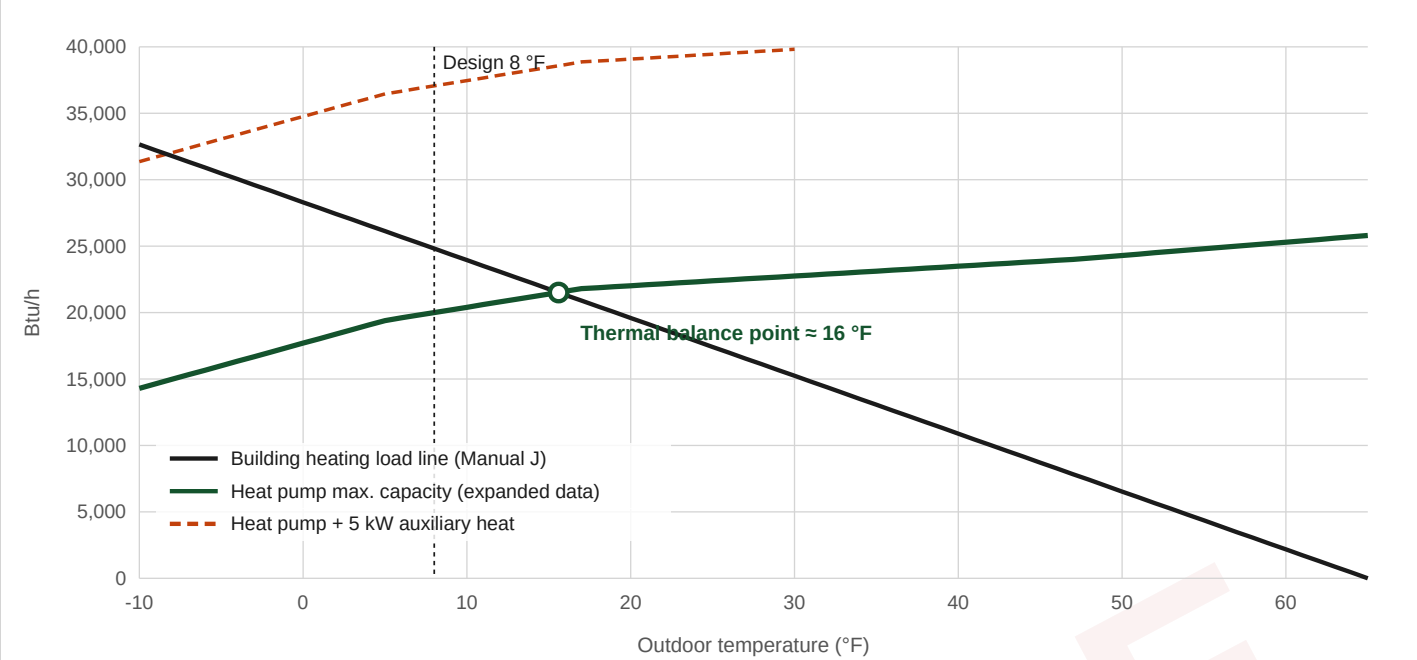
Cooling at design	Load Btu/h	Capacity Btu/h	Capacity / load
Total	16,031	20,000	125%
Sensible	12,711	14,800	116%
Latent	3,320	5,200	157%

Heating	Value
Heating load at 8 °F	24,819 Btu/h
Heat pump max. capacity at 8 °F (interpolated)	20,000 Btu/h · COP 2.23
Auxiliary heat required at design	4,819 Btu/h
Auxiliary heat selected	5.0 kW = 17,061 Btu/h · covers 69% of design load in emergency mode
Thermal balance point	15.6 °F (load line zero at 65 °F)
Supply air temperature at design, heat pump only	96 °F

Manual S check	Result	Status
Total cooling capacity 90%–130% of total cooling load	125%	**PASS**
Sensible capacity ≥ sensible load	116%	**PASS**
Latent capacity ≥ latent load	157%	**PASS**
Heat pump + auxiliary heat ≥ heating design load	149%	**PASS**
Auxiliary heat locked out above balance point (outdoor sensor)	16 °F	**PASS**

Window applied in this sample for a variable-capacity heat pump in a heating-dominated climate. Confirm the limits of the Manual S edition referenced by the AHJ.

SAMPLE



4 Duct design (Manual D)

Rigid round galvanized branch ducts, rectangular galvanized trunk 8" high, all in floor-truss space inside conditioned envelope; mastic-sealed.

Friction rate worksheet	in. w.c.
Blower external static pressure (manufacturer, at design cfm)	0.50
Filter, MERV 13, 4" media cabinet	– 0.12
Electric heat strip (5 kW)	– 0.04
Supply outlet (register / diffuser)	– 0.03
Return grille	– 0.03
Balancing damper	– 0.03
Available static pressure (ASP)	**0.25**

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FR = ASP × 100 / TEL = 0.25 × 100 / (111 + 80) = 0.131 in. w.c. per 100 ft
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TEL = measured length + equivalent length of fittings (supply 60 ft, return 50 ft per circuit). Shorter runs are trimmed with balancing dampers.

Supply branches

Run	Room	Htg cfm	Clg cfm	Design cfm	TEL ft	Size	Velocity fpm	FR actual
S1	Study	72	71	72	102	5" round	525	0.113
S2	Entry / Stair	49	20	49	88	5" round	356	0.054
S3	Mudroom / Laundry / Powder	72	36	72	78	5" round	531	0.115
S4	Living Room	118	115	118	99	6" round	601	0.117
S5	Kitchen / Dining	52	85	85	82	6" round	433	0.063
S6	Kitchen / Dining	52	85	85	78	6" round	433	0.063
S7	Bedroom 3	52	51	52	111	5" round	380	0.061
S8	Hall / Bath 2	33	29	33	98	4" round	374	0.078
S9	Bedroom 4	52	51	52	87	5" round	380	0.061
S10	Primary Bedroom	71	76	76	110	5" round	557	0.127
S11	Primary Bath	24	24	24	96	4" round	277	0.044
S12	Bedroom 2	53	56	56	89	5" round	411	0.071

SAMPLE

Supply trunk, returns and transfer air

Section	cfm	Size	Equiv. dia.	Velocity fpm	FR
Trunk A – plenum to x = 20 ft (all runs)	700	16" × 8"	12.2"	788	0.098
Trunk B – x = 20 ft to west end	314	8" × 8"	8.7"	706	0.113
RG-1 Central return, 1st floor	415	12" round, grille 16"×16"		529	
RG-2 Central return, 2nd floor hall	288	9" round, grille 14"×14"		651	

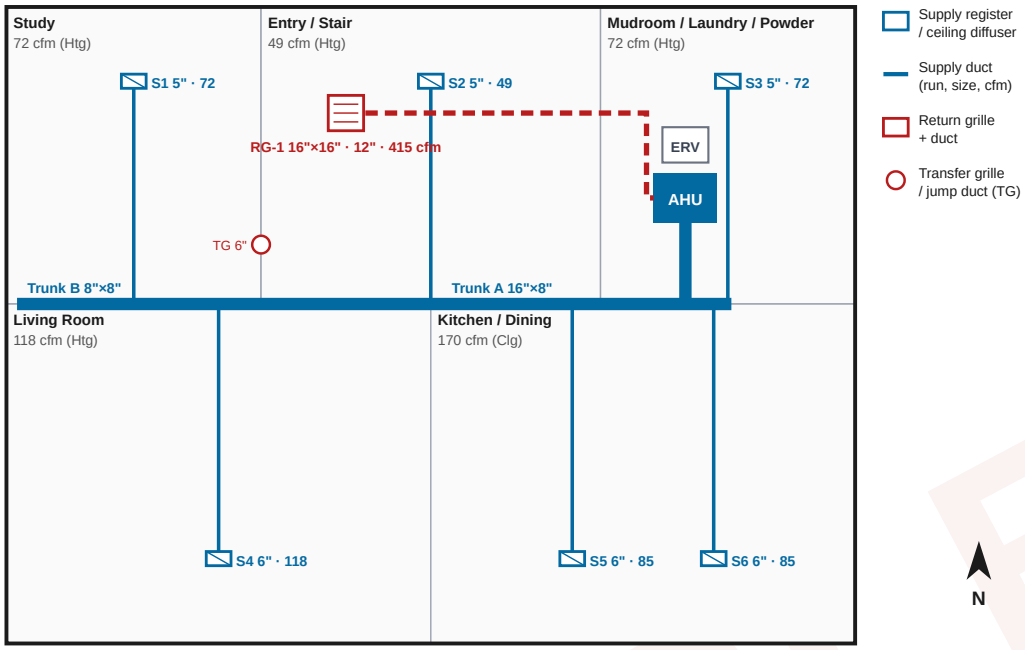
Room with door	Return path cfm	Transfer grille / jump duct
Study	72	jump duct 6" or transfer grille, ≤ 3 Pa room pressure with door closed
Bedroom 3	52	jump duct 6" or transfer grille, ≤ 3 Pa room pressure with door closed
Bedroom 4	52	jump duct 6" or transfer grille, ≤ 3 Pa room pressure with door closed
Primary Bedroom	76	jump duct 7" or transfer grille, ≤ 3 Pa room pressure with door closed
Bedroom 2	56	jump duct 6" or transfer grille, ≤ 3 Pa room pressure with door closed

Velocity limits applied: supply 900 fpm, return 700 fpm. Return grille sized for ≤ 400 fpm face velocity at 70 % free area.

SAMPLE

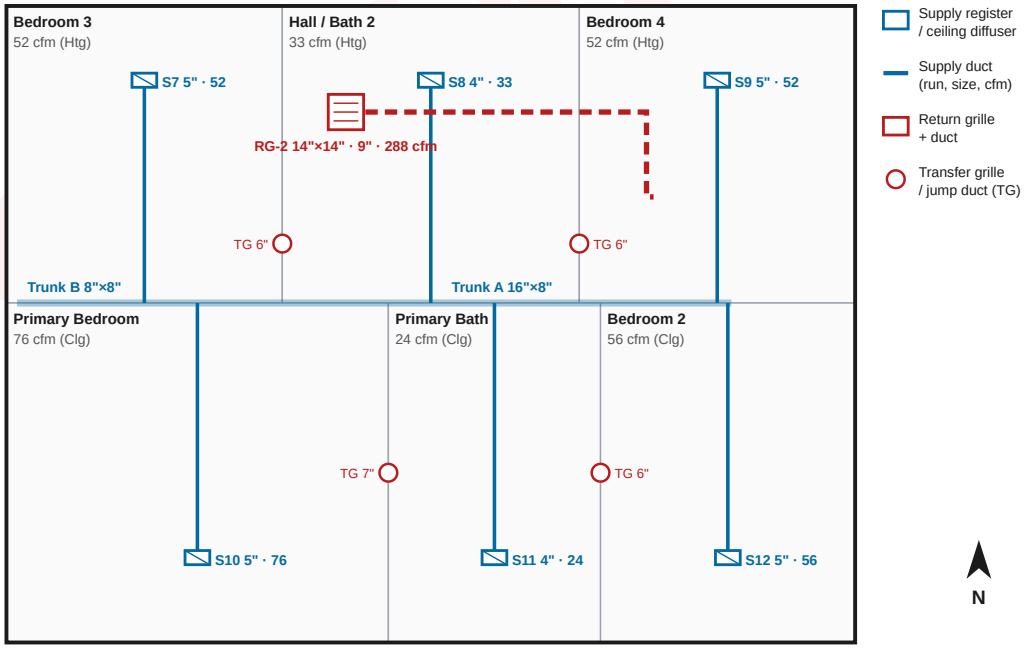
5 Duct layout (schematic)

First Floor – supply, return and transfer air



Scale: grid 1 ft = 12 px · overall 40' × 30' · drawing schematic, not to construction tolerance

Second Floor – supply, return and transfer air (trunk and AHU below, in floor-truss space / first floor)



Scale: grid 1 ft = 12 px · overall 40' × 30' · drawing schematic, not to construction tolerance

SAMPLE

6 Whole-dwelling ventilation (ASHRAE 62.2)

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Q_tot = 0.03 × CFA + 7.5 × (N_br + 1) = 0.03 × 2,400 + 7.5 × (4 + 1) = 72.0 + 37.5 = 109.5 cfm → 110 cfm continuous
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No infiltration credit taken (conservative). Balanced system: supply and exhaust airflows set equal at commissioning.

ERV supply to	cfm	ERV exhaust from	cfm
Living Room	25	Primary Bath	40
Study	20	Hall / Bath 2	30
Primary Bedroom	20	Mudroom / Laundry / Powder	40
Bedroom 2	15		
Bedroom 3	15		
Bedroom 4	15		
Total fresh air	**110**	**Total exhaust**	**110**

Space	Local exhaust
Kitchen	Range hood, 100 cfm intermittent, ducted to outside
Primary Bath	ERV exhaust 40 cfm continuous (meets 20 cfm continuous)
Bath 2	ERV exhaust 30 cfm continuous (meets 20 cfm continuous)
Powder Room	ERV exhaust 20 cfm continuous

7 Installation and commissioning notes

- Set blower to the design airflow in heating and cooling. Verify total airflow and static pressure at start-up and record readings.
- Balance each supply run to the design cfm within ±10 %. Balancing dampers at every takeoff, accessible.
- Seal all duct seams and connections with mastic. Duct leakage testing per IECC R403.3 as required by the AHJ.
- Install outdoor temperature sensor; lock out auxiliary heat above the balance point except during defrost.
- Commission the ERV to the scheduled supply and exhaust airflows and verify continuous operation.
- Any change to windows, insulation, air leakage target or equipment requires a revised calculation.

Basis and limitations

Load method follows the procedure structure of ACCA Manual J 8th Edition with simplified CLTD and peak solar factors; adjusted excursion (AED) and orientation checks are not performed in this sample. Equipment data are illustrative. This document is a design calculation, not a sealed engineering document. Where the AHJ requires a design professional's seal, the package is reviewed and sealed by a licensed professional engineer in the project state.

Designer: [Designer Name], [e.g. ACCA Manual J/D training · RESNET HERS Rater · ASHRAE BEMP]
[Company Name] – Remote HVAC & Energy Design · [email]

Signature / date

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